STACKS, DERIVED CATEGORIES OF SPACES: COMPUTING-SECTIONS-AS-COLIM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7022-7040 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 7042-7085, the complete author proof (42 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 49 of queue_spaceperfect.json (pb=2028, stmt 7022-7040, proof 7042-7085).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-computing-sections-as-colim' also exists in stacks-perfect — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 08IN: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4883–4892. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08IR: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5080–5095. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CRV: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6969–6980. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CRU: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6932–6944. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08FB: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 863–868. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08JJ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7085–7085. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07SK: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7085–7085. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08K0: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7085–7085. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let A be a ring. Let $f : U \to X$ be a flat morphism of algebraic spaces of finite presentation over A. Then*

(1) *there exists an inverse system of perfect objects L_n of $D(\mathcal{O}_X)$ such that*
$$R\Gamma(U, Lf^*K) = \text{hocolim } R\mathrm{Hom}_X(L_n, K)$$
in $D(A)$ functorially in K in $D_{QCoh}(\mathcal{O}_X)$, and

(2) *there exists a system of perfect objects E_n of $D(\mathcal{O}_X)$ such that*
$$R\Gamma(U, Lf^*K) = \text{hocolim } R\Gamma(X, E_n \otimes^{\mathbf{L}} K)$$
in $D(A)$ functorially in K in $D_{QCoh}(\mathcal{O}_X)$.

Proof. By Lemma 08IN we have
$$R\Gamma(U, Lf^*K) = R\Gamma(X, Rf_*\mathcal{O}_U \otimes^{\mathbf{L}} K)$$
functorially in K. Observe that $R\Gamma(X, -)$ commutes with homotopy colimits because it commutes with direct sums by Lemma 08FB. Similarly, $- \otimes^{\mathbf{L}} K$ commutes with derived colimits because $- \otimes^{\mathbf{L}} K$ commutes with direct sums (because direct sums in $D(\mathcal{O}_X)$ are given by direct sums of representing complexes). Hence to prove (2) it suffices to write $Rf_*\mathcal{O}_U = \text{hocolim} E_n$ for a system of perfect objects E_n of $D(\mathcal{O}_X)$. Once this is done we obtain (1) by setting $L_n = E_n^\vee$, see Cohomology on Sites, Lemma 08JJ.

Write $A = \operatorname{colim} A_i$ with A_i of finite type over $\mathbf{Z}$. By Limits of Spaces, Lemma 07SK we can find an i and morphisms $U_i \to X_i \to \mathrm{Spec}(A_i)$ of finite presentation whose base change to $\mathrm{Spec}(A)$ recovers $U \to X \to \mathrm{Spec}(A)$. After increasing i we may assume that $f_i : U_i \to X_i$ is flat, see Limits of Spaces, Lemma 08K0. By Lemma 08IR the derived pullback of $Rf_{i,*}\mathcal{O}_{U_i}$ by $g : X \to X_i$ is equal to $Rf_*\mathcal{O}_U$. Since Lg^* commutes with derived colimits, it suffices to prove what we want for f_i. Hence we may assume that U and X are of finite type over $\mathbf{Z}$.

Assume $f : U \to X$ is a morphism of algebraic spaces of finite type over $\mathbf{Z}$. To finish the proof we will show that $Rf_*\mathcal{O}_U$ is a homotopy colimit of perfect complexes. To see this we apply Lemma 0CRV. Thus it suffices to show that $R^i f_*\mathcal{O}_U$ has countable sets of sections over affines étale over X. This follows from Lemma 0CRU applied to the structure sheaf. □